

**Aim: To understand Continuous Integration, install and configure Jenkins with Maven/Ant/Gradle to setup a build Job**

### **Theory:**

Continuous Integration (CI) is a DevOps practice where code changes are automatically built, tested, and integrated into a shared repository multiple times a day. It helps in early detection of errors, reduces integration problems, and improves software quality.

### **Jenkins: An Overview**

Jenkins is an open-source CI/CD automation tool used for building, testing, and deploying applications. It allows developers to automate software development workflows and ensures a seamless integration process. Jenkins supports various build tools like **Maven**, **Ant**, and **Gradle** to compile and package applications.

### **Installing and Configuring Jenkins**

#### **1. Download and Install Jenkins**

- Install Java (JDK) as a prerequisite.
- Download Jenkins from the official website and install it on the server.
- Start Jenkins and configure initial setup using an administrator password.

#### **2. Installing Build Tools**

- Install **Maven**, **Ant**, or **Gradle** depending on project requirements.
- Configure Jenkins to recognize the installed build tool.

#### **3. Creating a Build Job in Jenkins**

- Navigate to **Jenkins Dashboard** → **New Item** → **Freestyle Project/Pipeline**.
- Configure the **Git repository URL** to fetch the source code.
- Select the **Build Tool (Maven/Ant/Gradle)** and define the build command.
- Set up triggers (e.g., Git webhooks) for automatic build execution.
- Save and trigger the build job to verify the setup.

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To install Jenkins following software packages are required:

- 1) GIT (git-scm.com)
- 2) Notepad++ (https://notepad-plus-plus.org/downloads/)
- 3) Latest Java development kit (JDK)
- 4) Jenkins
- 5) Apache Maven (Optional)

Step 1-: Install GIT

Step 2 -: Install Notepad++

Step 3 -: Install Java

Step 4 -: Install Jenkins

Step 5 -: Install Maven

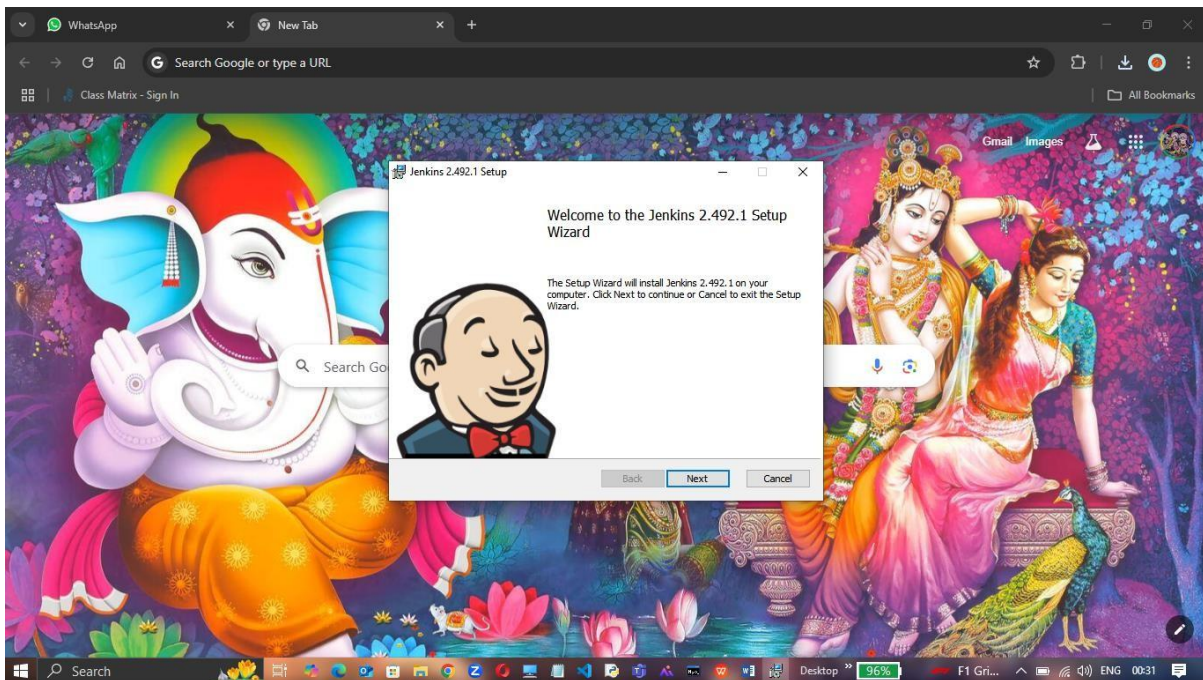
Jenkins is an open source automation tool written in Java with plugins built for Continuous Integration purpose. Jenkins is used to build and test your software projects continuously making it easier for developers to integrate changes to the project, and making it easier for users to obtain a fresh build. It also allows you to continuously deliver your software by integrating with a large number of testing and deployment technologies.

**Step 1-:** Open <https://www.jenkins.io/doc/book/installing/windows/> and install Jenkins.

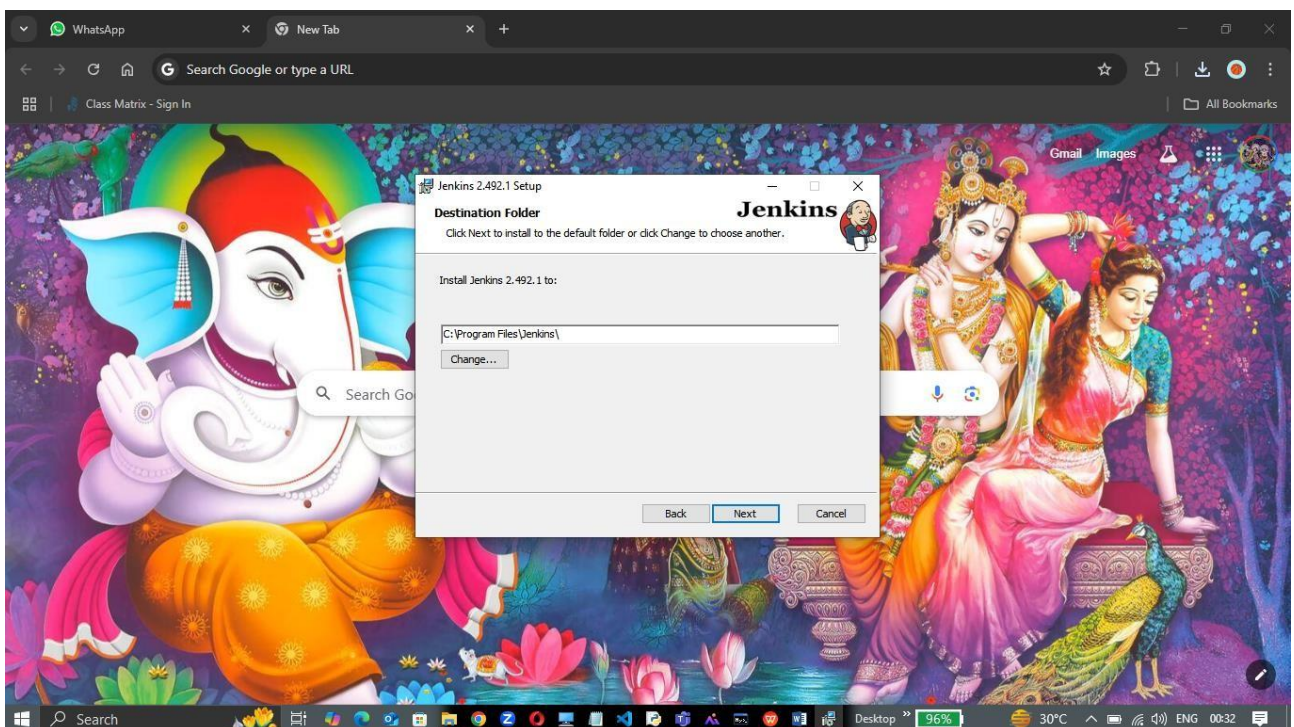
Open the installed .exe setup

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**Step 2:** Locate the folder where you want to install Jenkins in the location path:

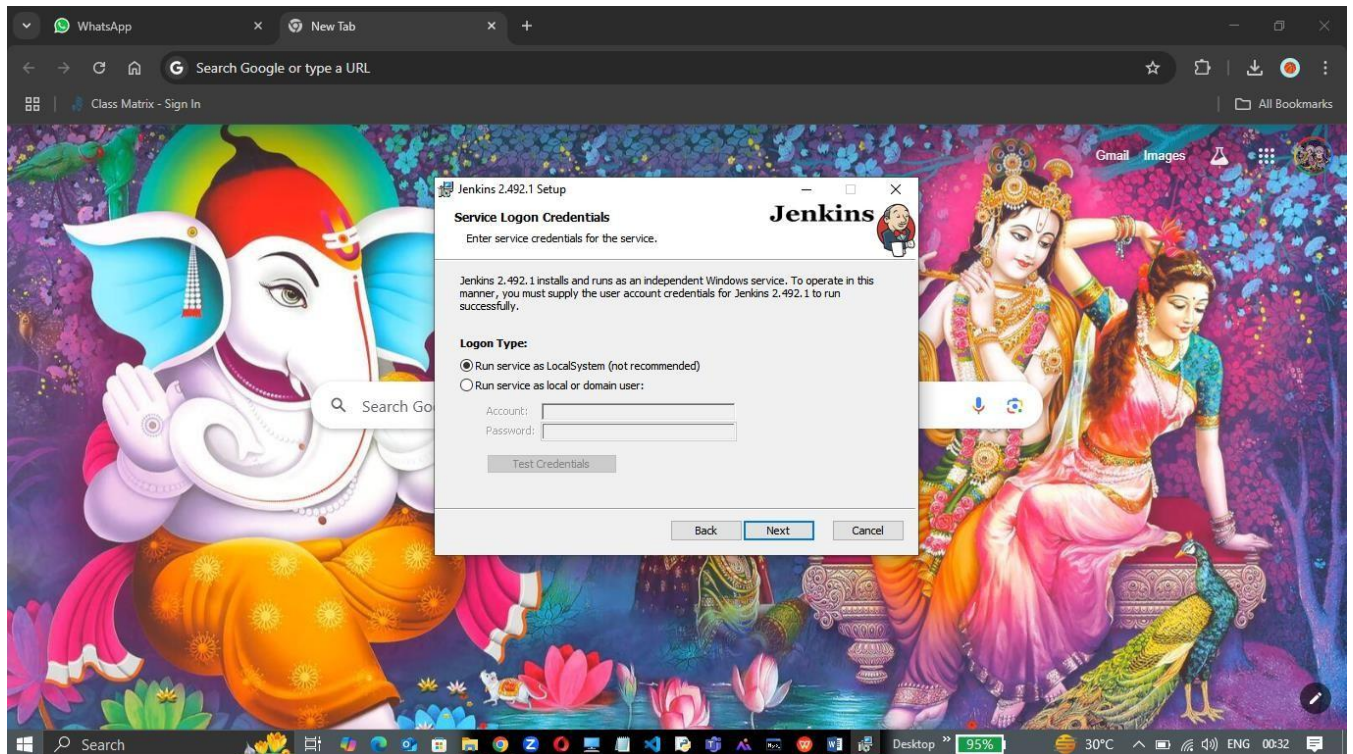




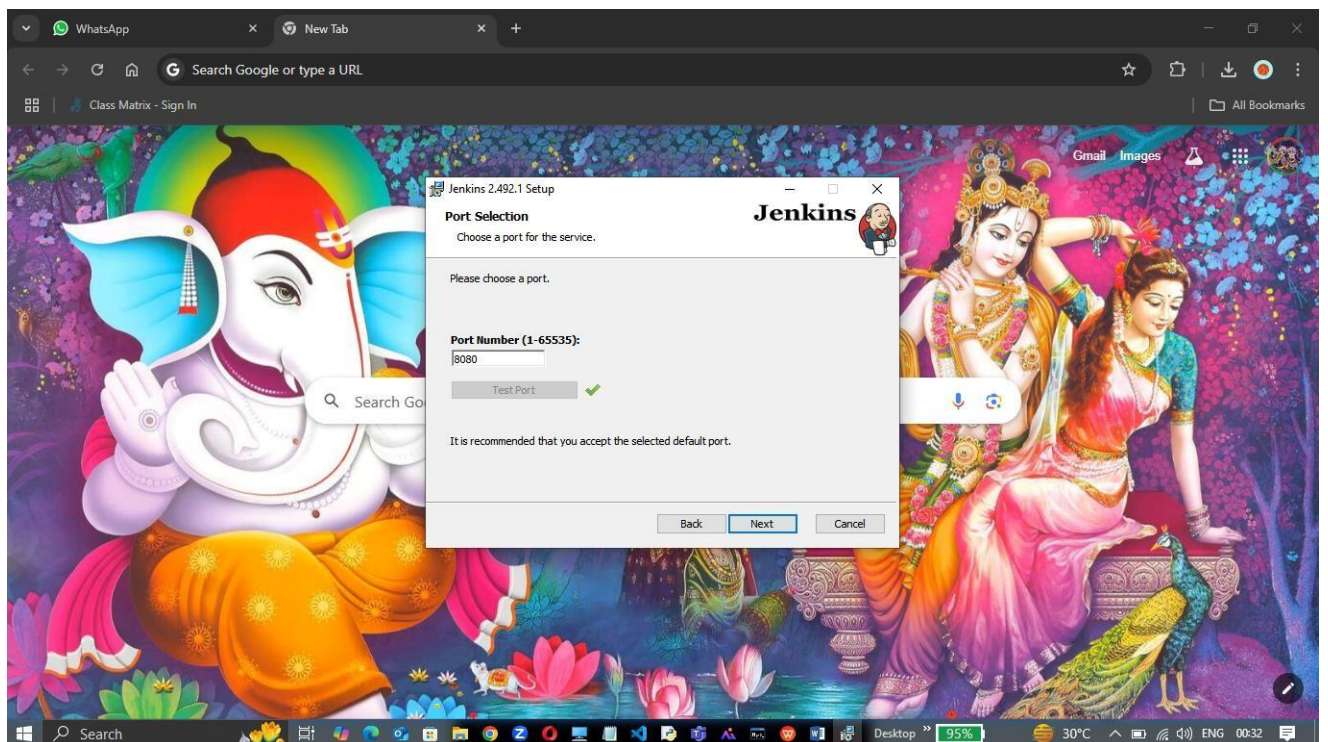
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**Step 3:** Select service as Local System and proceed to Next.



**Step 4:** Select the port 8080 and click Test Port button. The green tick will appear after which you can proceed to Next.

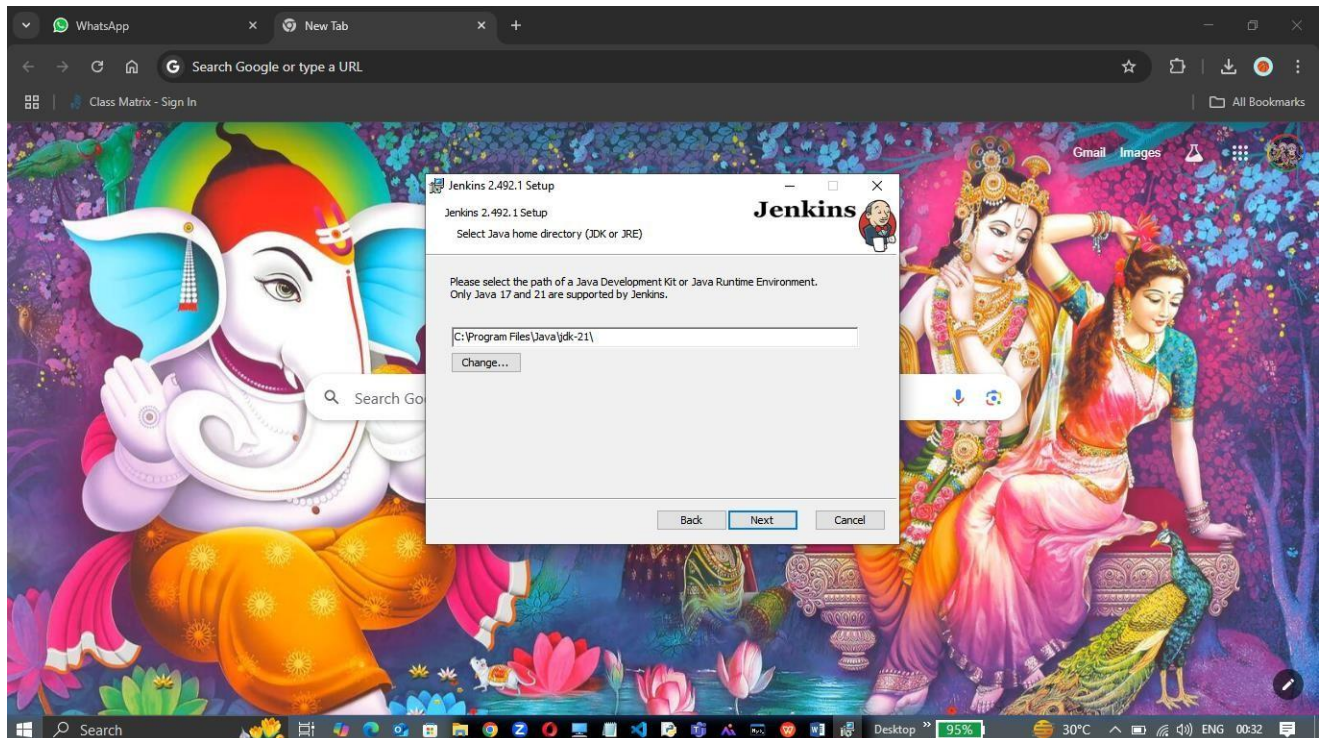




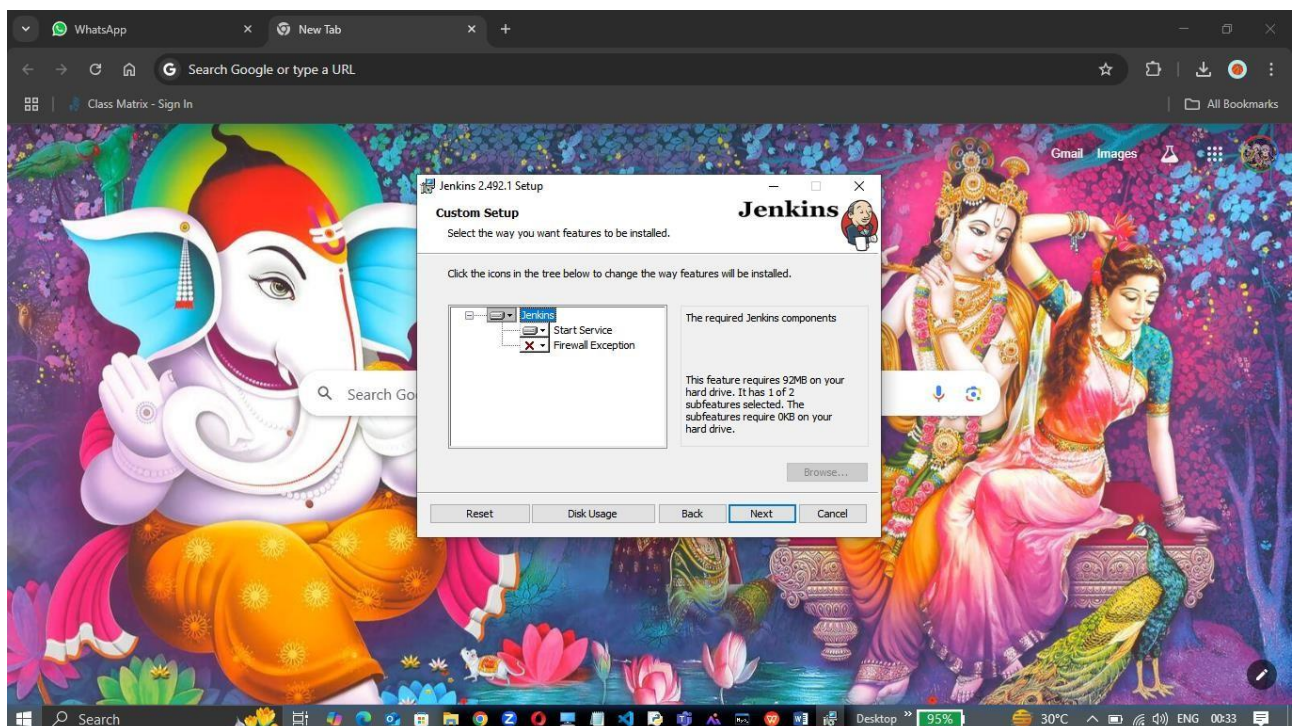
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**Step 5:** Locate the folder where you have installed JDK in the location path:



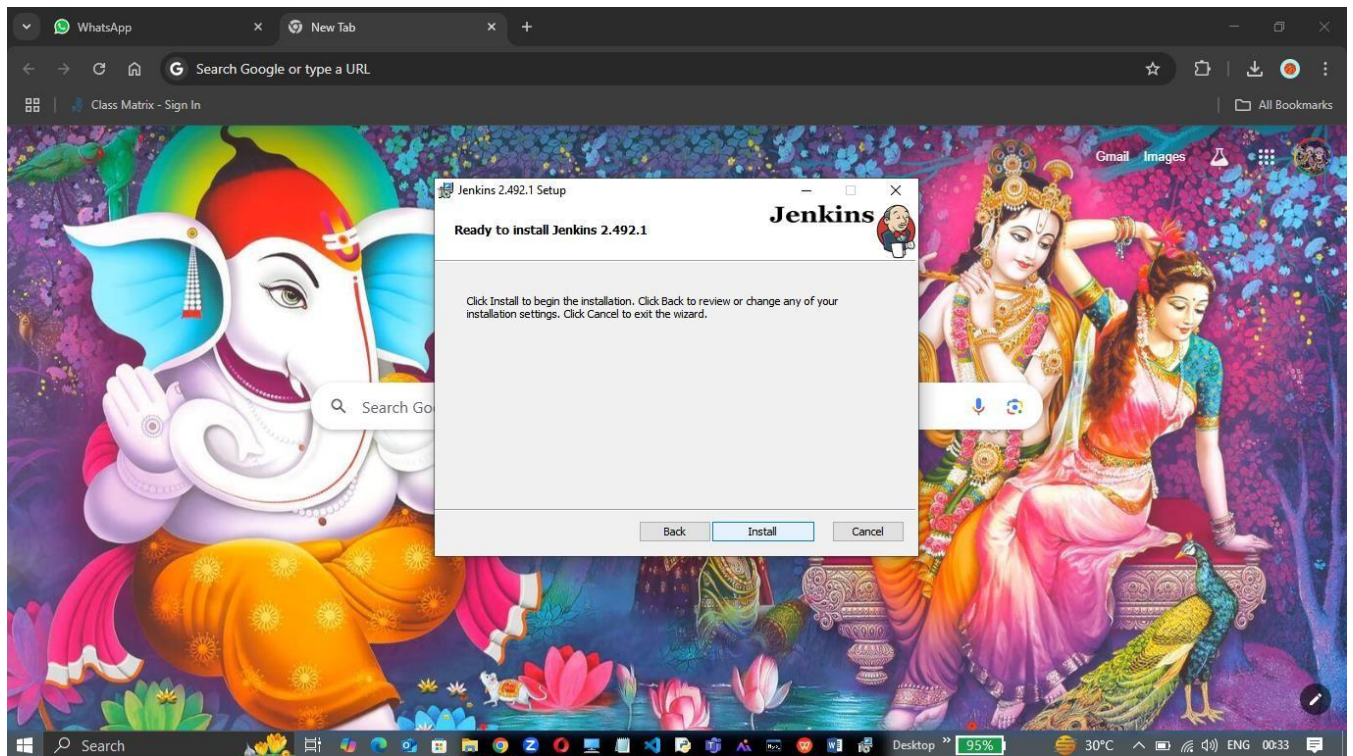
**Proceed to Next**



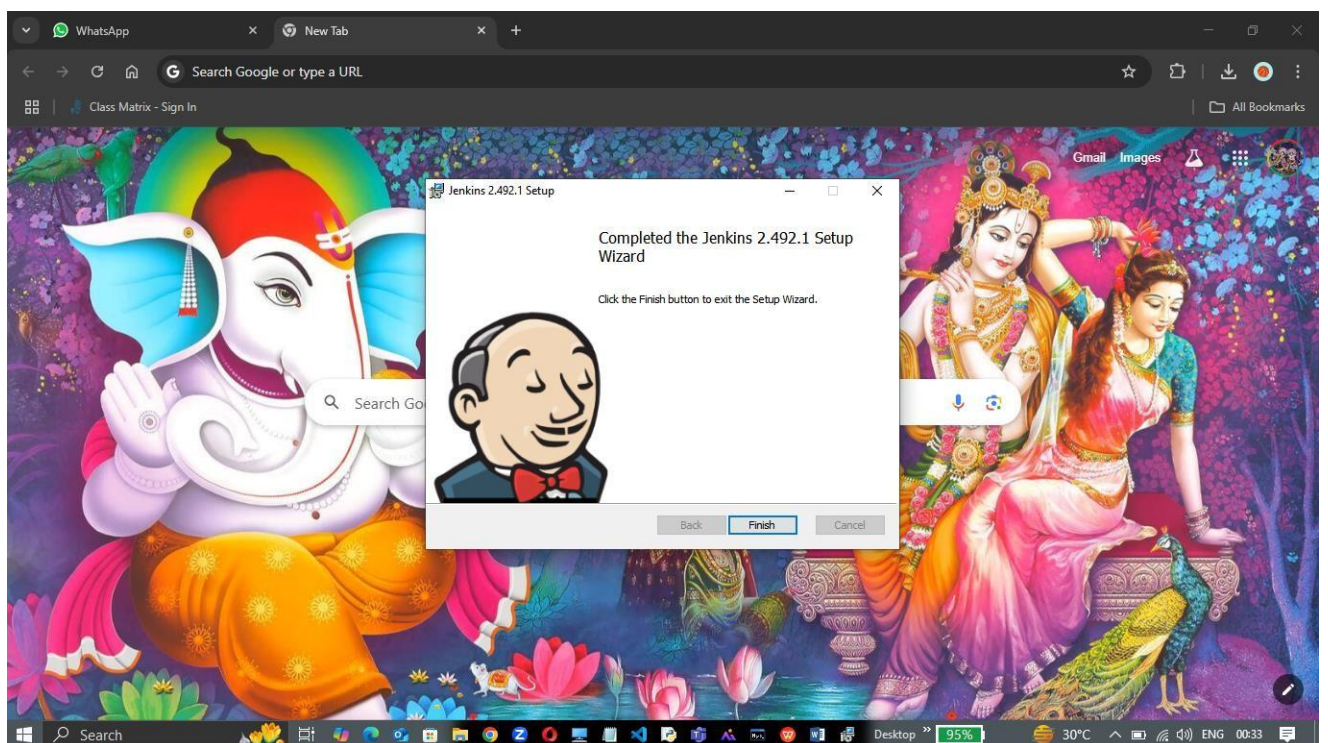


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**On clicking 'Install', installation is finished.**

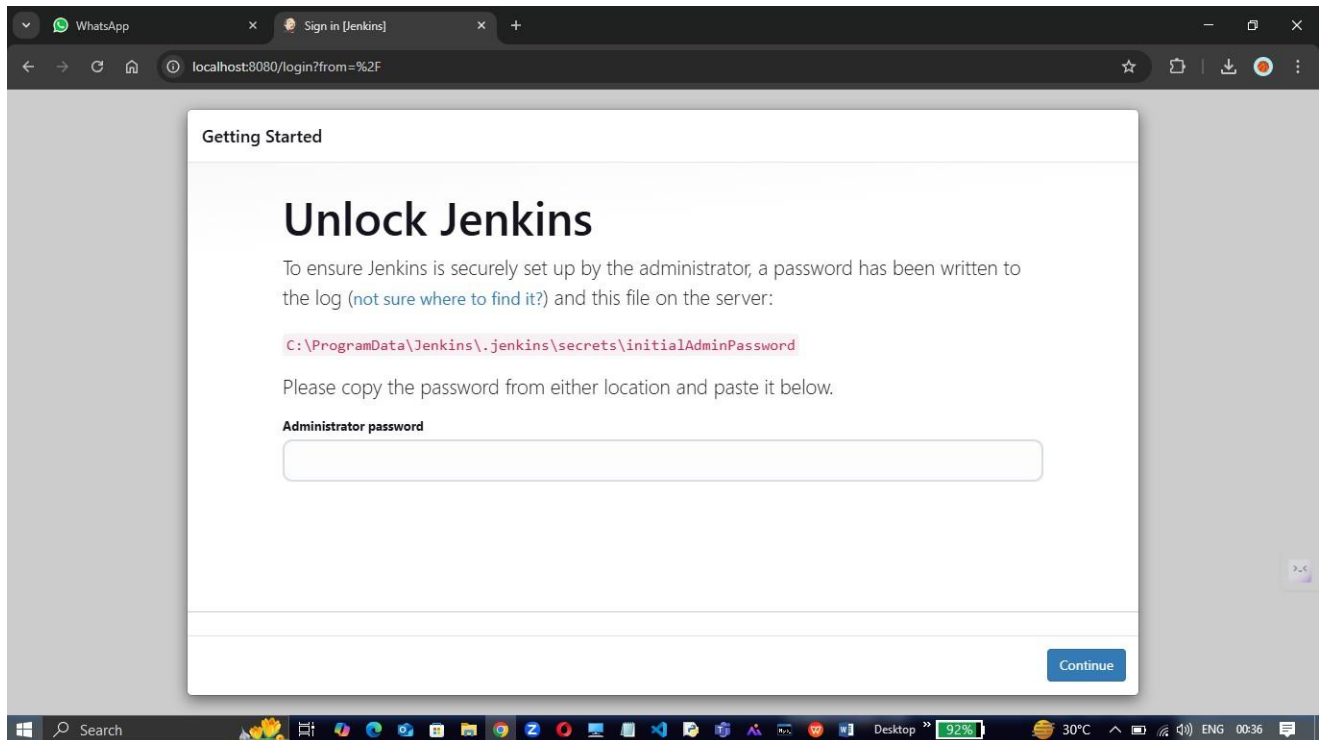


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**Step 6:** Once Installation is done, you can test the Jenkins on <http://localhost:8080> on the browser.

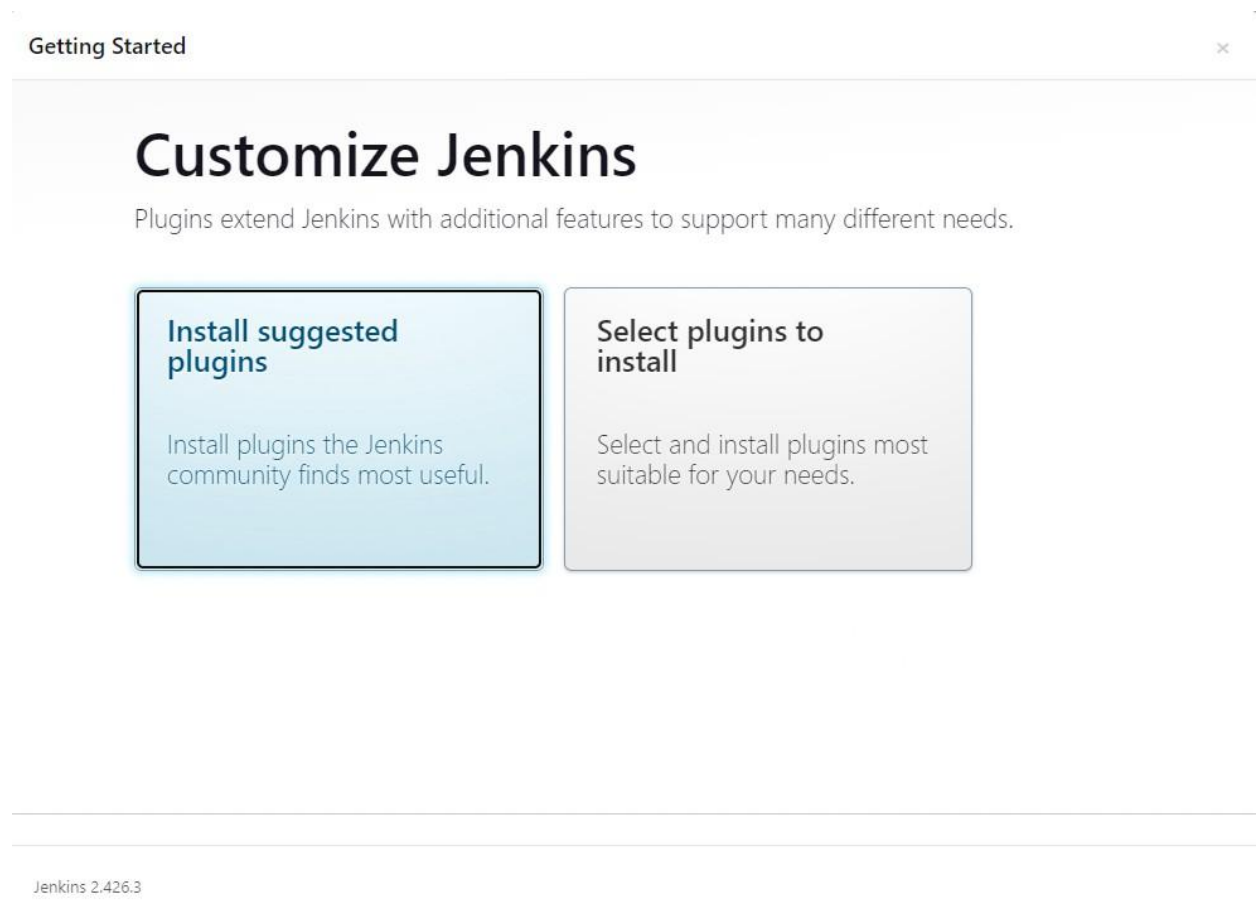
First time, when you open Jenkins portal it will ask to put admin default password which is stored in `/var/lib/jenkins/secrets/initialAdminPassword` file.



**Step 7:** On entering the password, you can continue to choose “Install Suggested Plugins”

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Once plugins are installed, click on next and specify the admin details along with the new password for Jenkins admin and click on finish to complete the installation.

After filling the details, click on Save & Continue, you will be redirected to the dashboard.



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### Getting Started

# Getting Started

|               |                          |                                   |                       |
|---------------|--------------------------|-----------------------------------|-----------------------|
| ✓ Folders     | ✓ OWASP Markup Formatter | ✓ Build Timeout                   | ✓ Credentials Binding |
| ✓ Timestamper | Workspace Cleanup        | Ant                               | Gradle                |
| Pipeline      | GitHub Branch Source     | Pipeline: GitHub Groovy Libraries | Pipeline: Stage View  |
| Git           | SSH Build Agents         | Matrix Authorization Strategy     | PAM Authentication    |
| LDAP          | Email Extension          | Mailer                            |                       |

REST API

\*\*\* bouncycastle API  
\*\*\* Instance Identity  
\*\*\* JavaBeans Activation Framework (JAF) API  
\*\*\* JavaMail API  
\*\*\* Credentials  
\*\*\* Plain Credentials  
\*\*\* Gson API  
\*\*\* Trilead API  
\*\*\* SSH Credentials  
\*\*\* Credentials Binding  
\*\*\* SCM API  
\*\*\* Pipeline: API  
\*\*\* commons-lang3 v3.x Jenkins API  
\*\*\* Timestamper  
\*\*\* Caffeine API  
\*\*\* Script Security  
\*\*\* JAXB  
\*\*\* SnakeYAML API  
\*\*\* Jackson 2 API  
\*\*\* commons-text API  
\*\*\* Pipeline: Supporting APIs  
\*\*\* Plugin Utilities API  
\*\*\* Font Awesome API  
\*\*\* Bootstrap 5 API  
\*\*\* JQuery3 API  
\*\*\* - required dependency

Jenkins 2.426.3

Dashboard

New Item

People

Build History

Manage Jenkins

My Views

Build Queue

No builds in the queue.

Build Executor Status

1 Idle

2 Idle

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job

Set up a distributed build

Set up an agent

Configure a cloud

Learn more about distributed builds

REST API

Jenkins 2.426.3

9

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Batch:-T21

Name & Roll No:- Meet Raut - 2201084

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Dashboard >

Enter an item name

» Required field



**Freestyle project**  
This is the central feature of Jenkins. Jenkins will build your project, combining any SCM with any build system, and this can be even used for something other than software build.



**Pipeline**  
Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.



**Multi-configuration project**  
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.



**Folder**  
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.



**Multibranch Pipeline**  
Creates a set of Pipeline projects according to detected branches in one SCM repository.



**Organization Folder**  
Creates a set of multibranch project subfolders by scanning for repositories.

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The screenshot shows the Jenkins Configuration page for a job named 'example 1'. The breadcrumb trail is 'Dashboard > example 1 > Configuration'. On the left, the 'Configure' section is active, with sub-items: General, Source Code Management, Build Triggers, Build Environment (selected), Build Steps, and Post-build Actions. The main area shows configuration options for 'example 1':

- ☐ Add timestamps to the Console Output
- ☐ Inspect build log for published build scans
- ☐ Terminate a build if it's stuck
- ☐ With Ant ?

Below these is the 'Build Steps' section. A step titled 'Execute Windows batch command ?' is present. The 'Command' field contains the text: `echo "hello tsec"`. A link 'See the list of available environment variables' is provided. Below the command field is an 'Advanced' dropdown menu. At the bottom of the configuration area are 'Save' and 'Apply' buttons.

The screenshot shows the Jenkins Console Output page for a job named 'example 1'. The breadcrumb trail is 'Dashboard > example 1 > #11 > Console Output'. On the left, the 'Console Output' section is active, with sub-items: Status, Changes, Console Output (selected), View as plain text, Edit Build Information, Delete build '#11', and Previous Build. The main area displays the console output for build #11, which is successful (indicated by a green checkmark icon). The output text is:

```
Started by user Muskan Tolani
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\example 1
[example 1] $ cmd /c call C:\Windows\TEMP\jenkins6203665954710491391.bat

C:\ProgramData\Jenkins\jenkins\workspace\example 1>echo "hello tsec"
"hello tsec"

C:\ProgramData\Jenkins\jenkins\workspace\example 1>exit 0
Finished: SUCCESS
```



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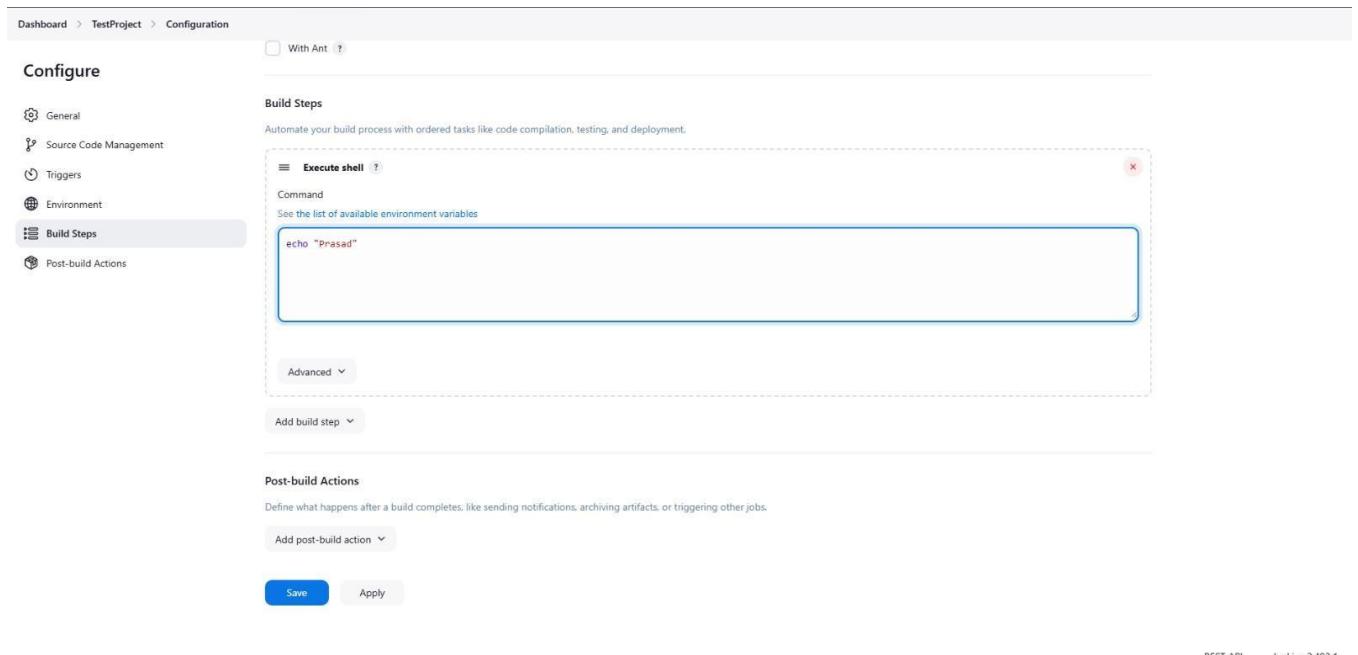
The screenshot shows the Jenkins Dashboard. On the left, there's a sidebar with links: New Item, Build History, Manage Jenkins, and My Views. Below these are two dropdowns: 'Build Queue' (showing 'No builds in the queue') and 'Build Executor Status' (showing '0/2'). The main area displays a table of build jobs. The table has columns for 'S' (Success), 'W' (Warning), 'Name', 'Last Success', 'Last Failure', and 'Last Duration'. The jobs listed are 'example-1', 'My\_example', 'myes', and 'r'. The 'example-1' job is successful, while the others are failed. At the bottom right, there's a 'REST API' link and the version 'Jenkins 2.492.1'.

| S | W | Name       | Last Success | Last Failure | Last Duration |
|---|---|------------|--------------|--------------|---------------|
| ✓ | ☀ | example-1  | 1 yr 1 mo #1 | N/A          | 0.17 sec      |
| ✗ | ☁ | My_example | N/A          | 2 yr 0 mo #5 | 19 ms         |
| ✗ | ☁ | myes       | 2 yr 0 mo #3 | 2 yr 0 mo #8 | 0.72 sec      |
| ✗ | ☁ | r          | 2 yr 0 mo #3 | 1 yr 1 mo #4 | 0.71 sec      |

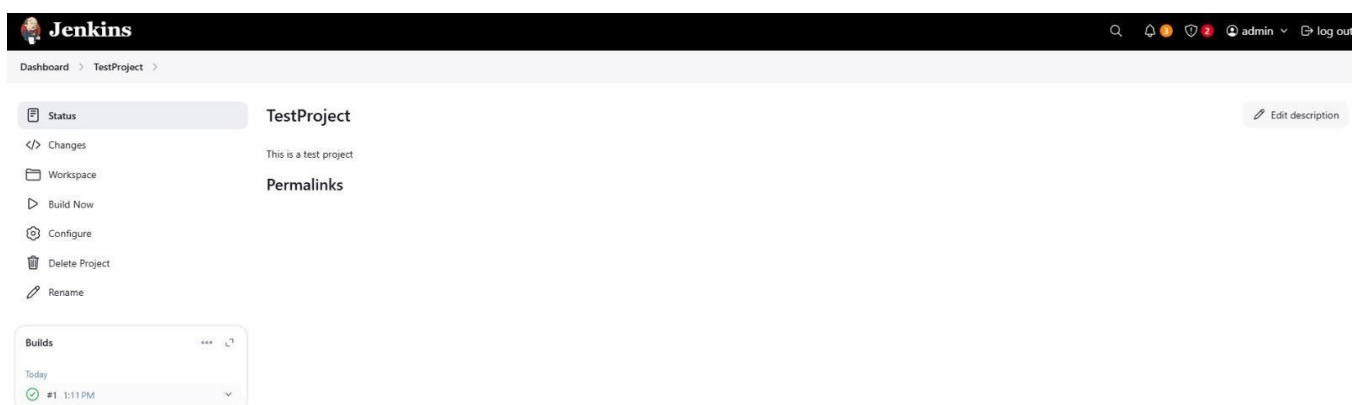
The screenshot shows the 'New Item' page in Jenkins. It has a header 'New Item' and a sub-header 'Enter an item name'. Below this is a text input field containing 'TestProject'. Underneath, there's a section 'Select an item type' with several options: 'Freestyle project' (Classic, general-purpose job type), 'Pipeline' (Orchestrates long-running activities), 'Multi-configuration project' (Suitable for projects that need a large number of different configurations), 'Folder' (Creates a container that stores nested items), 'Multibranch Pipeline' (Creates a set of Pipeline projects), and 'Organization Folder' (Creates a set of multibranch project subfolders). At the bottom, there's an 'OK' button.

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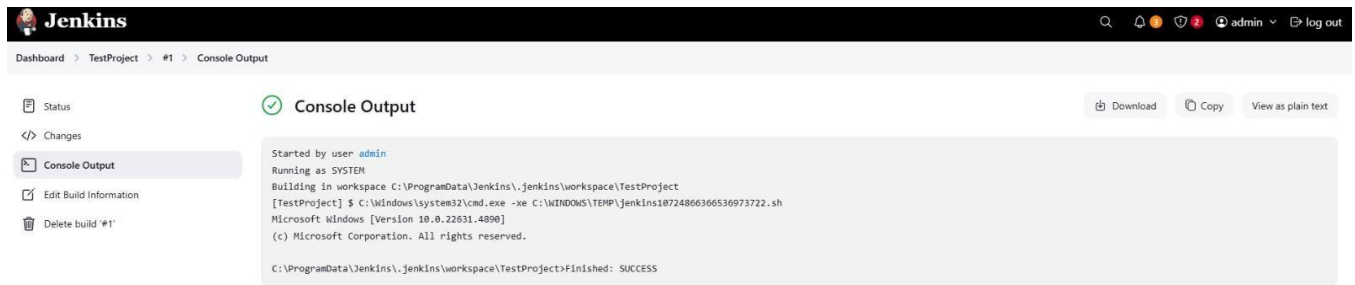
The screenshot shows the Jenkins Configuration page for a project named 'TestProject'. The breadcrumb navigation is 'Dashboard > TestProject > Configuration'. On the left, the 'Configure' section is active, with a sidebar listing options: General, Source Code Management, Triggers, Environment, Build Steps (selected), and Post-build Actions. The main area is titled 'Build Steps' and includes a checkbox for 'With Ant'. Below this, a 'Build Steps' section allows automating the build process with ordered tasks. A step named 'Execute shell' is added, with a command field containing 'echo "Prasad"'. An 'Advanced' dropdown is visible below the command field. At the bottom, there is an 'Add build step' button. The 'Post-build Actions' section is also present, with an 'Add post-build action' button. 'Save' and 'Apply' buttons are at the bottom right of the configuration area. The top right corner shows 'REST API' and 'Jenkins 2.492.1'.



The screenshot shows the Jenkins overview page for 'TestProject'. The breadcrumb navigation is 'Dashboard > TestProject'. The left sidebar contains links: Status (selected), Changes, Workspace, Build Now, Configure, Delete Project, and Rename. The main content area shows 'TestProject' with the description 'This is a test project'. Below this is a 'Permalinks' section. At the bottom left, a 'Builds' section shows a list of builds for 'Today', with the first build (#1) completed at 1:11 PM. The top right of the page shows the Jenkins logo, a search icon, and user information 'admin' with a 'log out' link.

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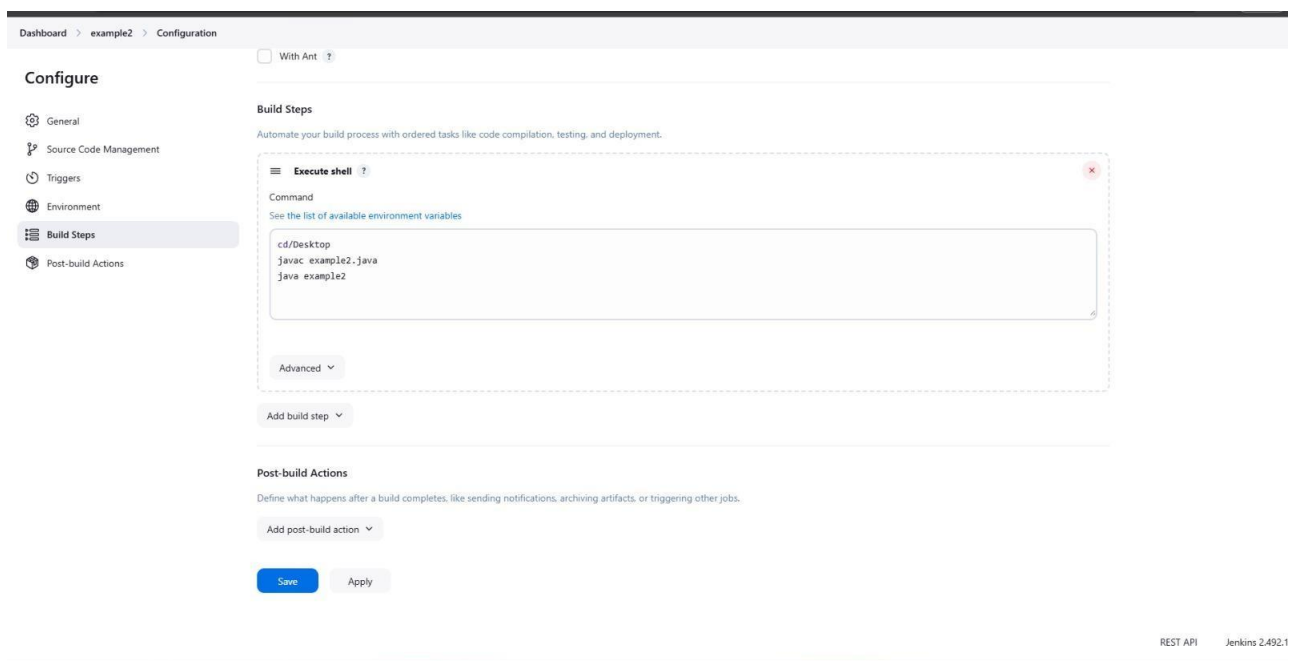
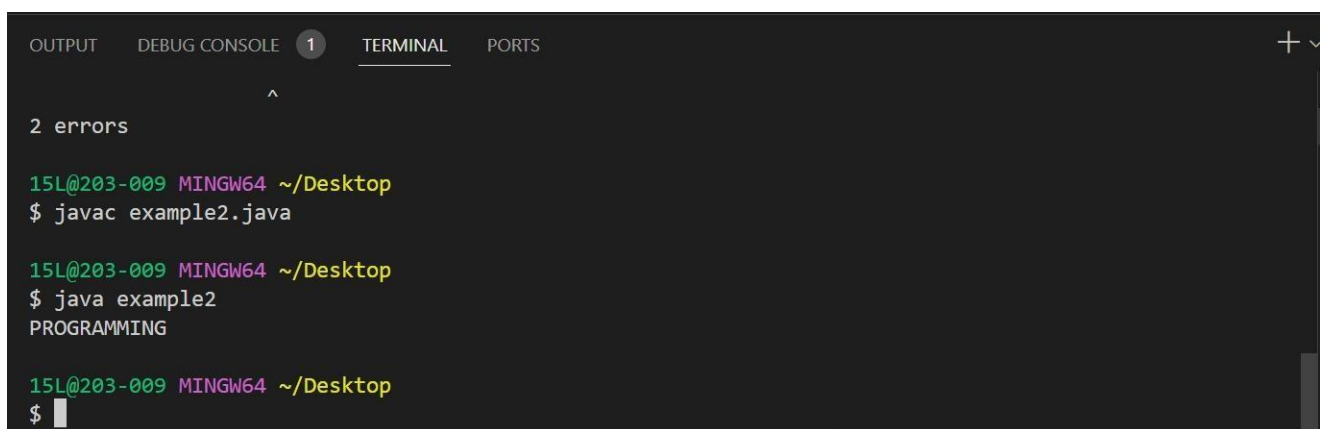
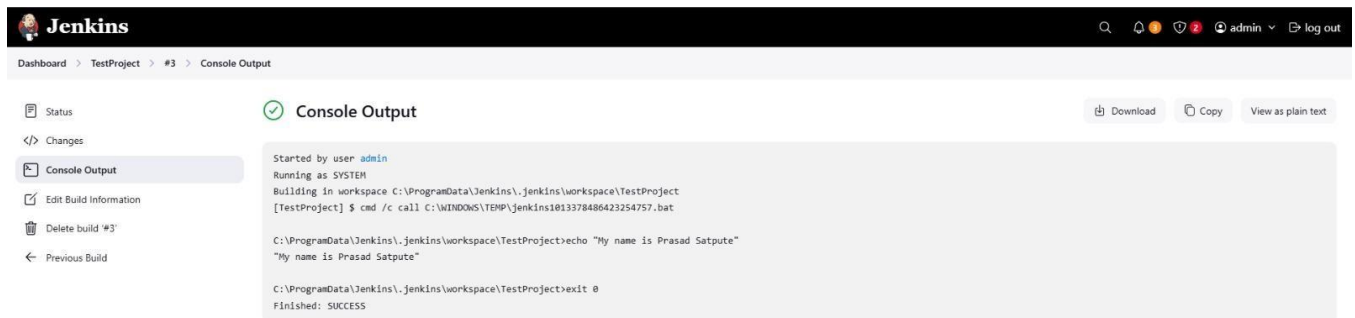
```
15L@203-009 MINGW64 ~  
$ cat > example1.sh  
#!/bin/bash  
name=$1  
Address=$2  
echo "Hello $name ..your address is $Address"  
  
[1]+  Stopped                  cat > example1.sh  
  
15L@203-009 MINGW64 ~  
$
```

```
15L@203-009 MINGW64 ~  
$ bash example1.sh  
Hello ..your address is  
  
15L@203-009 MINGW64 ~  
$ bash example1.sh Prasad  
Hello Prasad ..your address is  
  
15L@203-009 MINGW64 ~  
$ bash example1.sh Prasad Santacruz  
Hello Prasad ..your address is Santacruz
```



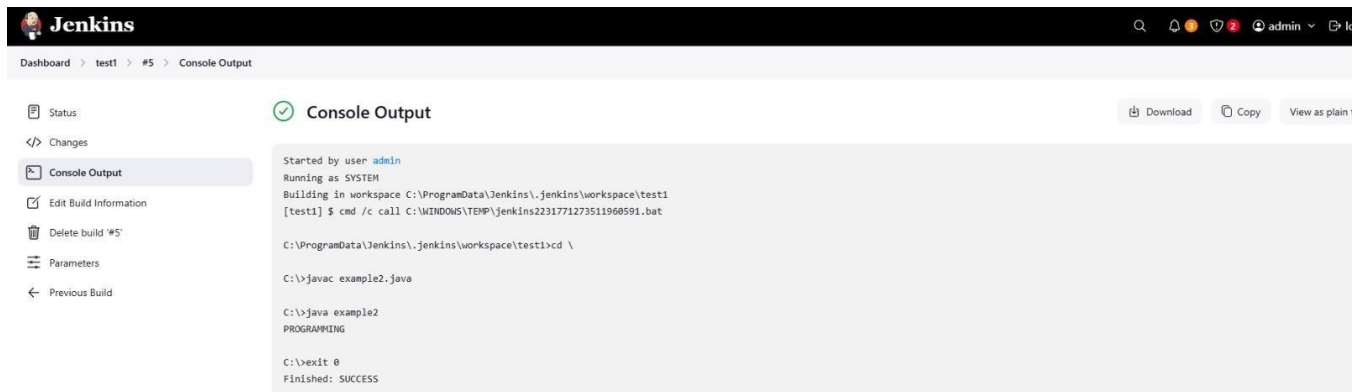
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The screenshot shows the Jenkins web interface for build #5 of job 'test1'. The left sidebar contains links for Status, Changes, Console Output (selected), Edit Build Information, Delete build #5, Parameters, and Previous Build. The main area displays the console output for build #5, which is successful. The output text is as follows:

```
Started by user admin
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\test1
[test1] $ cmd /c call C:\WINDOWS\TEMP\jenkins2231771273511960591.bat

C:\ProgramData\Jenkins\jenkins\workspace\test1>cd \

C:\>javac example2.java

C:\>java example2
PROGRAMMING

C:\>exit 0
Finished: SUCCESS
```



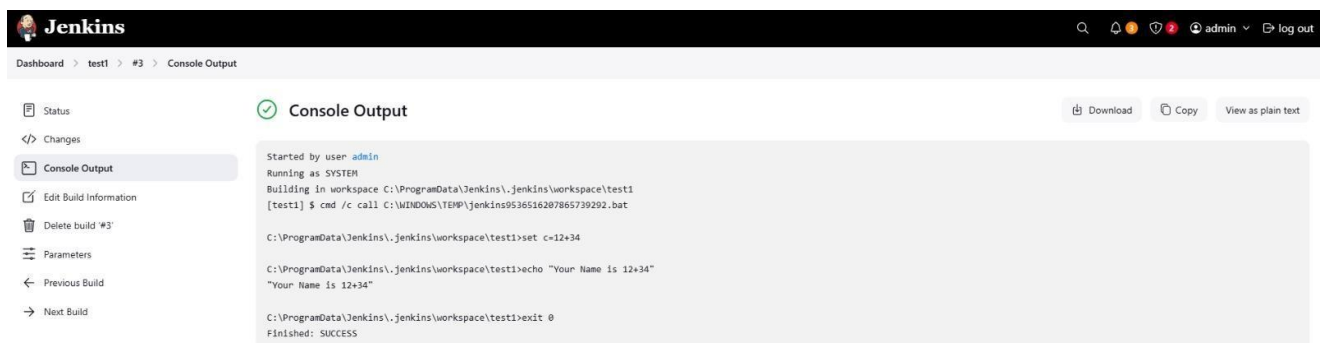
The screenshot shows the Jenkins web interface for build #4 of job 'test1'. The left sidebar contains links for Status, Changes, Console Output (selected), Edit Build Information, Delete build #4, Parameters, Previous Build, and Next Build. The main area displays the console output for build #4, which is successful. The output text is as follows:

```
Started by user admin
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\test1
[test1] $ cmd /c call C:\WINDOWS\TEMP\jenkins11493019808206271570.bat

C:\ProgramData\Jenkins\jenkins\workspace\test1>set /a c=1+2

C:\ProgramData\Jenkins\jenkins\workspace\test1>echo "Your Name is 3"
"Your Name is 3"

C:\ProgramData\Jenkins\jenkins\workspace\test1>exit 0
Finished: SUCCESS
```



The screenshot shows the Jenkins web interface for build #3 of job 'test1'. The left sidebar contains links for Status, Changes, Console Output (selected), Edit Build Information, Delete build #3, Parameters, Previous Build, and Next Build. The main area displays the console output for build #3, which is successful. The output text is as follows:

```
Started by user admin
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\test1
[test1] $ cmd /c call C:\WINDOWS\TEMP\jenkins9536516207865739292.bat

C:\ProgramData\Jenkins\jenkins\workspace\test1>set c=12+34

C:\ProgramData\Jenkins\jenkins\workspace\test1>echo "Your Name is 12+34"
"Your Name is 12+34"

C:\ProgramData\Jenkins\jenkins\workspace\test1>exit 0
Finished: SUCCESS
```

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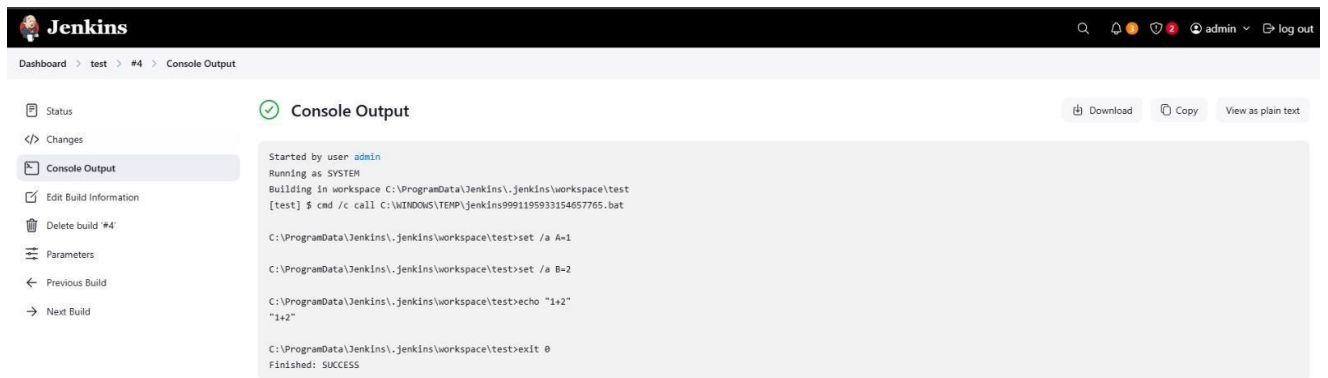


The screenshot shows the Jenkins web interface for build #2 of the 'test1' job. The left sidebar contains navigation links: Status, Changes, Console Output (selected), Edit Build Information, Delete build #2, Parameters, Previous Build, and Next Build. The main area displays the console output for build #2, which is successful. The output text is as follows:

```
Started by user admin
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\test1
[test1] $ cmd /c call C:\WINDOWS\TEMP\jenkins3591631450106967559.bat

C:\ProgramData\Jenkins\jenkins\workspace\test1>echo "Your Name is Sachin"
"Your Name is Sachin"

C:\ProgramData\Jenkins\jenkins\workspace\test1>exit 0
Finished: SUCCESS
```



The screenshot shows the Jenkins web interface for build #4 of the 'test' job. The left sidebar contains navigation links: Status, Changes, Console Output (selected), Edit Build Information, Delete build #4, Parameters, Previous Build, and Next Build. The main area displays the console output for build #4, which is successful. The output text is as follows:

```
Started by user admin
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\test
[test] $ cmd /c call C:\WINDOWS\TEMP\jenkins9991195933154657765.bat

C:\ProgramData\Jenkins\jenkins\workspace\test>set /a A=1

C:\ProgramData\Jenkins\jenkins\workspace\test>set /a B=2

C:\ProgramData\Jenkins\jenkins\workspace\test>echo "i+2"
"i+2"

C:\ProgramData\Jenkins\jenkins\workspace\test>exit 0
Finished: SUCCESS
```



The screenshot shows the Jenkins web interface for build #3 of the 'test' job. The left sidebar contains navigation links: Status, Changes, Console Output (selected), Edit Build Information, Delete build #3, Parameters, Previous Build, and Next Build. The main area displays the console output for build #3, which is successful. The output text is as follows:

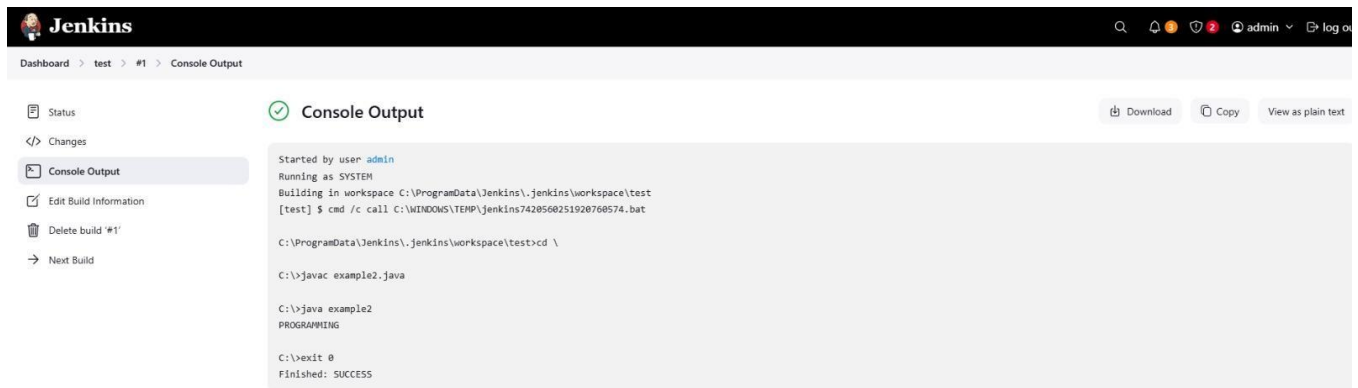
```
Started by user admin
Running as SYSTEM
Building in workspace C:\ProgramData\Jenkins\jenkins\workspace\test
[test] $ cmd /c call C:\WINDOWS\TEMP\jenkins2360247137534955462.bat

C:\ProgramData\Jenkins\jenkins\workspace\test>echo "ABC and DEF"
"ABC and DEF"

C:\ProgramData\Jenkins\jenkins\workspace\test>exit 0
Finished: SUCCESS
```

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**Conclusion:** Thus, we have successfully installed and configured Jenkins with Maven/Ant/Gradle to setup a build Job and learnt about the implementation of Jenkins in open source continuous integration.